



liftinstituut
SINCE 1933



EU-TYPE EXAMINATION CERTIFICATE

Issued by Liftinstituut B.V.
identification number Notified Body 0400,
commissioned by Decree no. 2022-0000107366

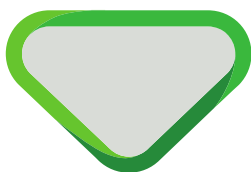
Certificate no.	: NL17-400-1002-262-02	Revision no.:	1
Description of the product	: Hydraulic valve for a brake element as part of a protection means against unintended downwards car movement		
Trademark	: Blain Hydraulics GmbH		
Type no.	: L20 3/4"		
Name and address of the manufacturer	: Blain Hydraulics GmbH Pfaffenstrasse 1 74078 Heilbronn Germany		
Name and address of the certificate holder	: Blain Hydraulics GmbH Pfaffenstrasse 1 74078 Heilbronn Germany		
Certificate issued on the following requirements	: Lifts Directive 2014/33/EU		
Certificate based on the following standard	: Parts of: EN 81-20:2020 EN 81-50:2020		
Test laboratory	: None		
Date and number of the laboratory report	: None		
Date of EU-type examination	: 19-07-2022		
Additional document with this certificate	: Annex belonging to the EU-type examination certificate no.: NL17-400-1002-262-02 rev.1		
Additional remarks	: See chapter 1 and 2 of the annex belonging to this EU- type examination certificate. This revision replaces certificate NL17-400-1002-262-02 of 10-10-2017		
Conclusion	: The safety component meets the requirements of the Lifts Directive 2014/33/EU taking into account any additional remarks mentioned above.		

Amsterdam

Date : 20-07-2022
Valid until : 20-07-2027


ing A.J. van Ommen
International Business
Manager


Certification decision by



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Annex of EU-type examination certificate NL17-400-1002-262-02

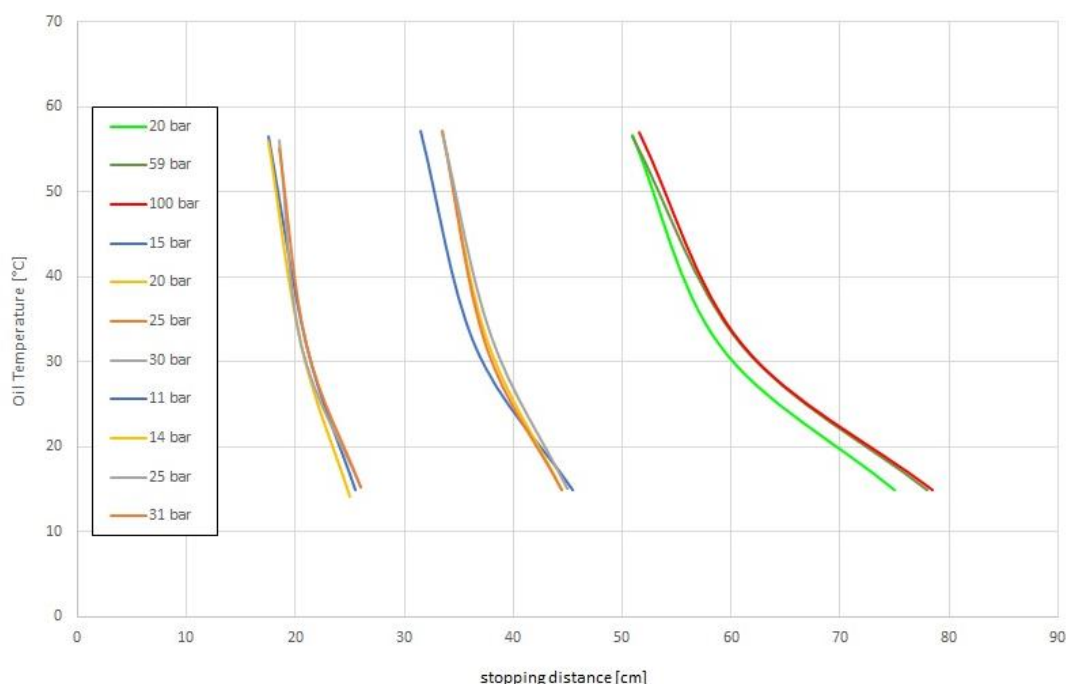
Date of original certificate : 10-10-2017
Revision number / date : 1 / 20-07-2022
Project number : P210326

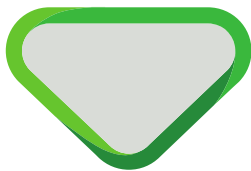
1. Description

The L20 Pressure Lock Valve is a solenoid operated check valve designed for hydraulic elevators and includes a self-closing manual lowering valve. Its purpose is to allow free flow of oil from the pump unit to the cylinder for upward travel and to prevent flow in the reverse direction from the cylinder to pump until an electrical signal is given to its solenoid. The L20 can be mounted directly on a Blain EV $\frac{3}{4}$ " valve at the cylinder port without the need of any adapters. It can be either used as a standalone safety valve by docking on an existing Blain EV valve with separate tank line (for modernizing old installations) or can be used as an integrated safety valve together with Blain's EVL valve without the need of separate tank connection.

Installed in the main cylinder line directly integrated with the main elevator control valve, the L20 can be employed as a safety back up valve to the down system of the main control valve to prevent unwanted down movement of the elevator should an electrical or mechanical malfunction occur in the main control valve (UCM case).

The stopping distance depends on the hydraulic pressure in the system, related to the diameter of the cylinder and the oil viscosity. Tests have been conducted with the maximum flow to find out the stopping distances are within 800 mm and the retardation did not exceed **1g** at various oil temperatures namely, 15°C, 32°C and 56°C. This resulted in the following graph for 3 types of cylinders (left is Ø85 mm, center is Ø63 mm, right is Ø50mm):





L20 is available in a Standard version and in a version with an optional filter FH.

GENERAL DATA & LIMITS

Type	L20
Max breaking distance after triggering	800 mm
Max permissible distance between level and triggering UCM detector	150 mm
Maximum nominal speed	1,0 m/s
Maximum release speed	1,3 m/s
Permissible nominal flow $\frac{3}{4}$ "	≤ 125 l/min
Nominal pressure range	8 – 100 bar
Viscosity range	20 cSt – 200 cSt
Temperature range	15 °C - 56 °C for ISO VG46

COIL

Applicable Voltages AC	24, 42, 110, 208/230 V
Applicable Voltages DC	12, 24, 42/48/60, 80, 100/125, 180/196 V

APPLICABLE VALVES

Type	EV 0, EV1, EV10, EV100
Valve port size	$\frac{3}{4}$ "

APPLICABLE SOLENOID VALVE SEAT SIZES

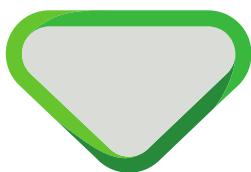
size	2,0 mm till nominal pressure of 50 bar
size	1,6 mm from nominal pressure above 50 bar

See annex 1 for a general overview of the product.

2. Conditions

Additional to, or in deviation of the applicable demands in the considered requirements / standards (see certificate and/or page 1 of this report), the following conditions shall be taken into account:

- The above mentioned safety component represents only one part of the protective equipment against movements of the car in downward direction. Only in combination with a detection and triggering component (also two different components are possible), which must be subjected to an own EU type-examination according to the test procedure specified in EN 81-50:2020, clause 5.8, the total system can fulfil the requirements for a UCM protection means in accordance with EN 81-20:2020, clause 5.6.7.
- The stopping distance and the allowable maximum deceleration is guaranteed within the limits of the system, set in chapter 2 of this report. Therefore separate certificates for braking distances and decelerations at different volume flows, different pressures or different viscosity is not required, as long as the total system is tested on site. This can be done at reduced speed.
- Maximum travelled distance before the UCM is detected shall be 150mm in order to meet the requirements in clause 5.6.7 of EN 81-20:2020.
- The installer shall provide suitable test instructions.



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- Both the A3 coil as the Down coil shall be energized according to clause 5.6.6.5 of EN 81-20:2020.
- The L20 coil is energized 300-500 msec in advance of energizing the main control valve (D coil) and the L20 coil is de-energized 300-500 msec delayed after main control valve (D coil) is de-energized, by normal movement. In this way the L20 valve does not take part in the normal operation of the lift and has not be monitored for proper opening and closing. In case of UCM both coils shall be de-energized simultaneously according to clause 5.6.7.8 of EN 81-20:2020.
- In case there is a risk of unintended car movement in the upward direction, appropriate measures shall be taken by the installer.
- Maintenance instructions shall be present on site and shall be met.

3. Conclusions

Based upon the results of the EU-type examination Liftinstituut B.V. issues an EU-type examination certificate.

The EU-type examination certificate is only valid for products which are in conformity with the same specifications as the type certified product. The EU-type examination certificate is issued based on the requirements that are valid at the date of issue. In case of changes of the product specifications, changes in the requirements or changes in the state of the art, the certificate holder shall request Liftinstituut B.V. to reconsider the validity of the EU-type examination certificate.

4. CE marking and EU Declaration of conformity

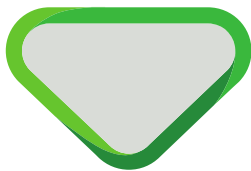
Every product that is placed on the market in complete conformity with the examined type must be provided with a CE marking according to art. 18 of the Lift directive 2014/33/EU under consideration that conformity with eventually other applicable Directives is proven. Also every product must be accompanied by an EU declaration of conformity according to annex II of the Directive in which the name, address and the Notified Body identification number of Liftinstituut B.V. shall be included as well as the number of the EU type-examination certificate.

An EU-type certified safety component shall be random checked, for example according to annex IX of the Lift directive 2014/33/EU before these safety components may be CE-marked and may be placed on the market. For further information on random checking by Liftinstituut, see regulation 2.0.1 'Regulations for product certification' on www.liftinstituut.com.

Prepared by:

W. Visser
Certification Expert

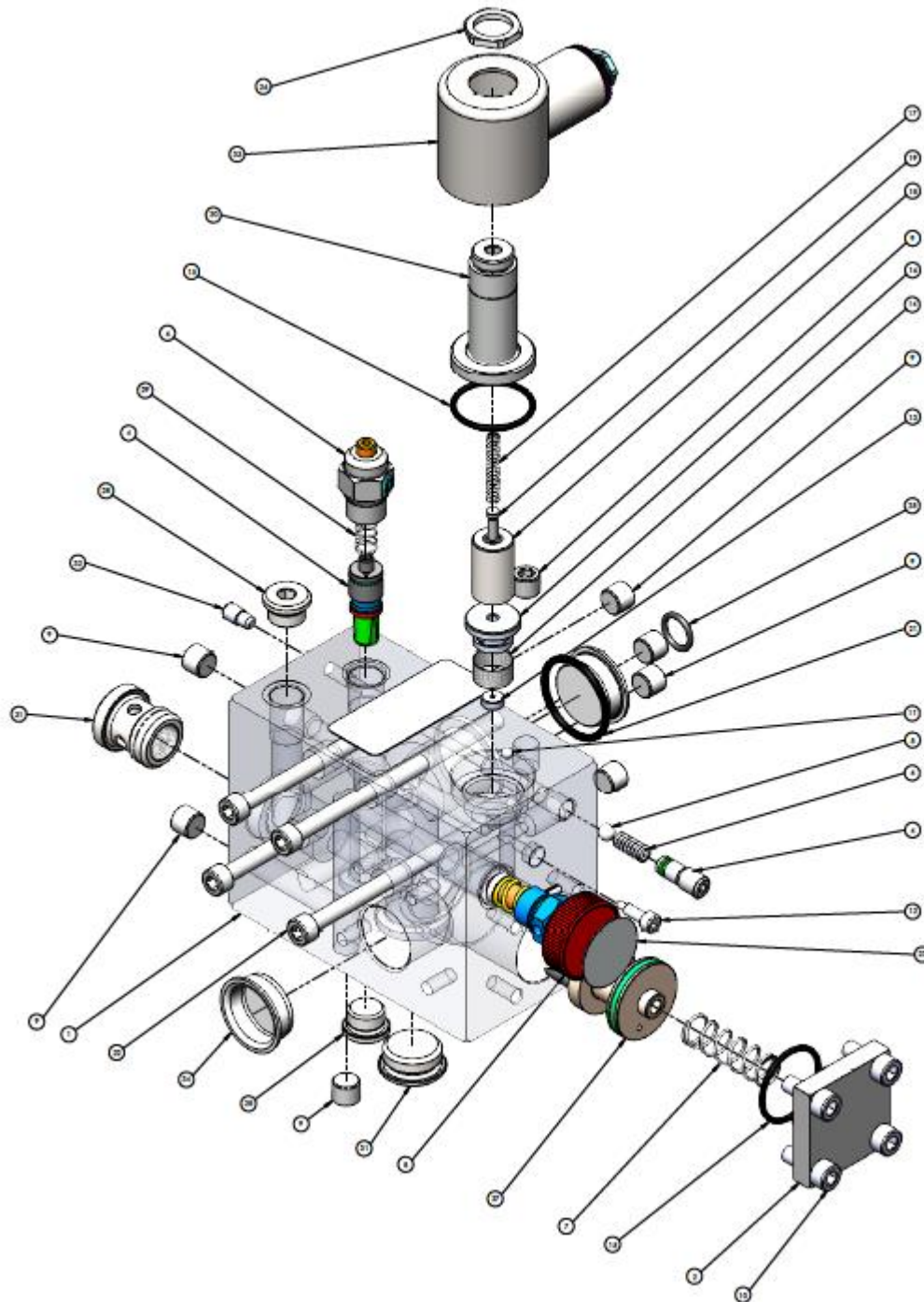
Certification decision by:

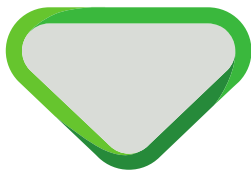


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Annex 1a. L20 with filter FH

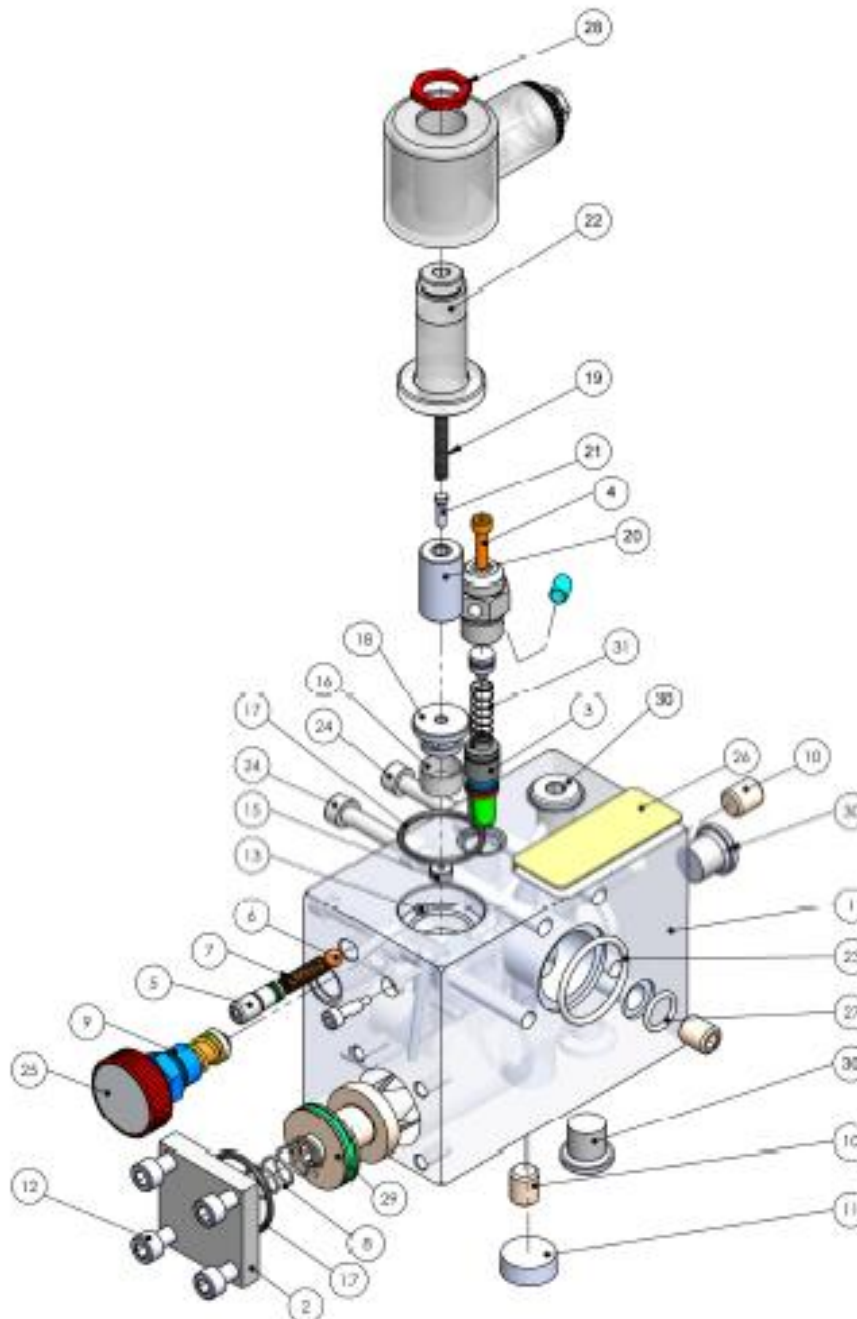


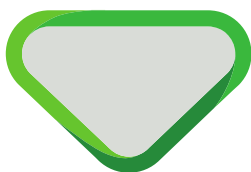


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Annex 1b. L20 Standard

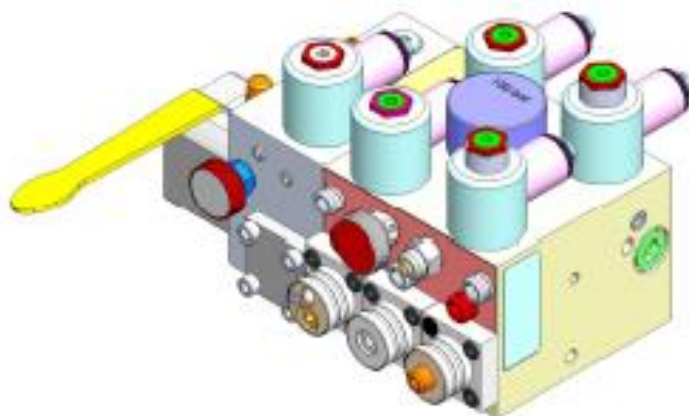




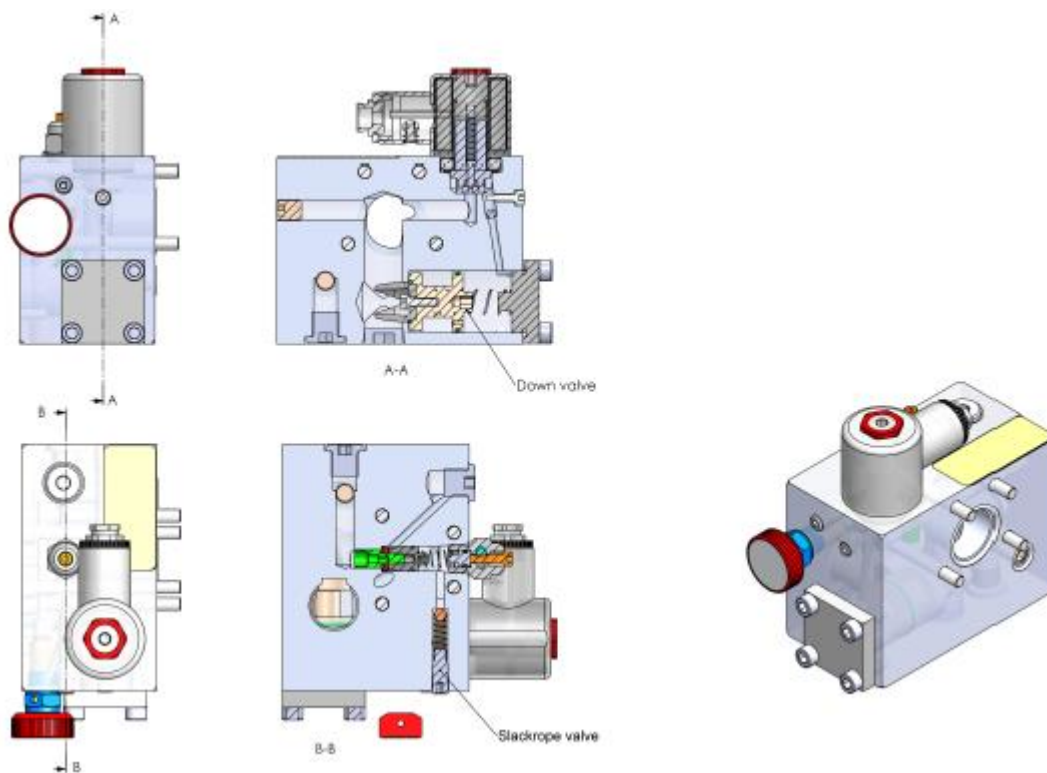
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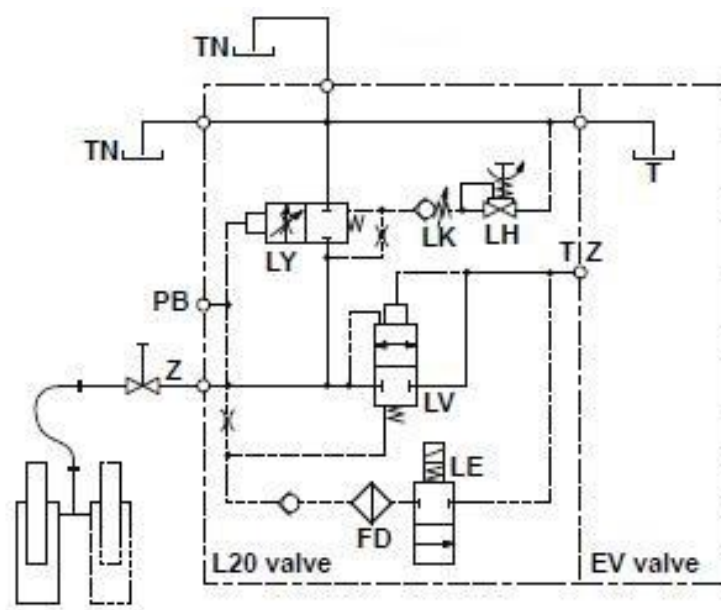
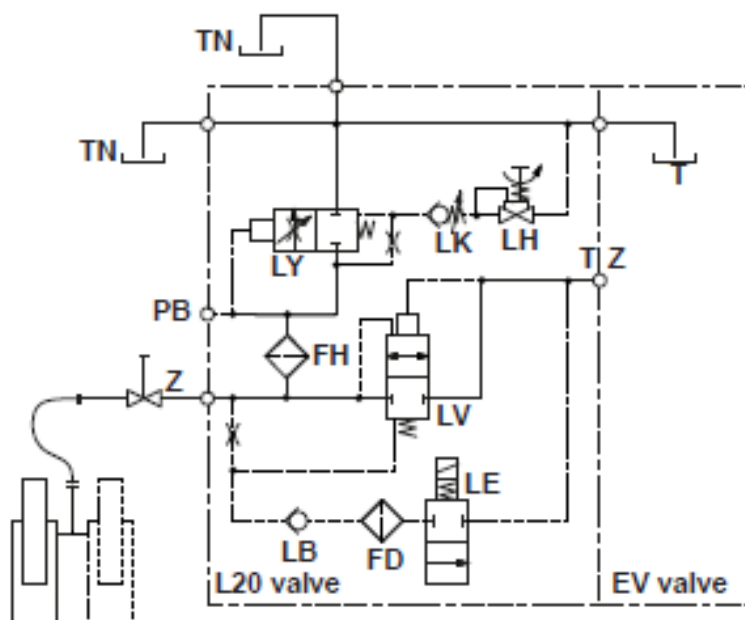


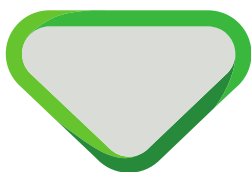
Annex 1c. Gen L20 connected to EV valve block 3/4 "



Annex 1d. L20 sectional views







Annex 2. Documents of the Technical File which were subject of the examination

Title	Document number	Date
L20 assembly with EV100	500405 / L 200 a	09-06-2016
L20 sectional views	500405 / L 200 a	09-06-2016
L20 Messungen Liftinstituut	Ver.1.0	21-06-2017
FH Filter drawing	500955 / L 10 Filter für 1.5"- 2.5"; L20	01-06-2022
L20 Parts list (BoM)	500405 / L 200 a	13-07-2022
Risk Analysis L20	Ver.2.0	13-07-2022
L20 User manual	July 2022	13-07-2022
F4.66 Test report	NL17-400-1002-262-02 rev.1	19-07-2022

Annex 3. Reviewed deviations from the standards

EN xx-x par.	Requirement	Accepted design
X.X.X		

Annex 4. Revision of the certificate and its annex

Rev.:	Date	Summary of revision
-	10-10-2017	Original
1	20-07-2022	- Renewal of the certificate with another 5 years - Updated to EN 81-20:2020 and EN 81-50:2020 - Added optional filter FH